

Session_8

Q.1 Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

A.1

```
#include <stdio.h>

void getUserInitials(char name[], char initials[]) {
    int i = 0;
    int j = 0; // First initial

    initials[j++] = name[0]; // Find space and take the next character

    while (name[i] != '\0') { if (name[i] == ' ') {
        initials[j++] = name[i + 1];
    }
    i++;
    }
    initials[j] = '\0';
}

int main() {
    char name[] = "Virat Kohli";
    char initials[10];

    getUserInitials(name, initials);

    printf("Initials: %s\n", initials);
    return 0;
}
```

Q.2 Create a function `addToCart` that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function. Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.

A.2 #include <stdio.h>

#include <string.h>

```
void addToCart(char cart[][50], int *count, char product[]) {
```

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```
        strcpy(cart[*count], product);
        (*count)++;

printf("\nUpdated Cart:\n");

for (int i = 0; i < *count; i++) {
printf("%d. %s\n", i + 1, cart[i]);
}
}

int main() {
char cart[10][50];
int count = 0;

addToCart(cart, &count, "Laptop");
addToCart(cart, &count, "Headphones");
addToCart(cart, &count, "Mouse");
return 0;
}
```

Q.3 Write two functions: `increaseFollowersByValue` and `increaseFollowersByReference`. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

A.3 `#include <stdio.h>`

```
// Pass-by-value
void increaseFollowersByValue(int followers) {
followers = followers + 1000;

printf("Inside pass-by-value: %d\n", followers);
}

// Using pointer (reference-like behavior)
void increaseFollowersByReference(int *followers) {
    *followers = *followers + 1000;
```

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```
    printf("Inside pass-by-reference: %d\n", *followers);
}
int main() {
    int followers = 5000;
    printf("Original followers: %d\n", followers);
    increaseFollowersByValue(followers);
    printf("After pass-by-value: %d\n", followers);
    increaseFollowersByReference(&followers);
    printf("After pass-by-reference: %d\n", followers);
    return 0;
}
```

Q.4 Build a function `formatPrice` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

A.4 `#include <stdio.h>`

```
void formatPrice(int price, char result[]) {
    int i = 0;
    int digits[20];
    int count = 0;
```

```
    // Store digits in reverse order
```

```
        while (price > 0) {
            digits[count++] = price % 10;
            price = price / 10;
        }
```

```
    result[i++] = 0xE2;
```

```
    result[i++] = 0x82;
```

```
    result[i++] = 0xB9;
```

```
    // Print digits in reverse
```

```
    for (int j = count - 1; j >= 0; j--) {
        result[i++] = digits[j] + '0';
    }
    result[i] = '\0';
```

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```
}  
int main() {  
    char price1[50];  
    char price2[50];  
  
    char price3[50];  
  
    formatPrice(1599, price1);  
    formatPrice(24999, price2);  
    formatPrice(999, price3);  
  
    printf("Laptop: %s\n", price1);  
    printf("Headphones: %s\n", price2);  
    printf("Mouse: %s\n", price3);  
    return 0;  
}
```

Q.5 Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string). Constraint: The refactored function should work for any string input, not just a specific use case.

A.5 #include <stdio.h>

```
void capitalizeFirstLetter(char text[]) {  
    if (text[0] >= 'a' && text[0] <= 'z') {  
        text[0] = text[0] - 32;  
    }  
}
```

```
int main() {  
    char product[] = "laptop";  
    char username[] = "virat";  
    capitalizeFirstLetter(product);  
    capitalizeFirstLetter(username);  
  
    printf("Product: %s\n", product);  
    printf("Username: %s\n", username);  
}
```

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```
return 0;  
}
```
